

Question:

The y -intercepts, P and Q , of two perpendicular lines intersecting at the point $A(6,8)$ have a sum of zero. What is the area of $\triangle APQ$?

- (A) 45 (B) 48 (C) 54 (D) 60 (E) 72

Reasoning + Answer:

<think>

1. Identify the given information:

- Lines are perpendicular and intersect at point $A(6,8)$.
- P and Q are the y -intercepts of the lines.
- P and Q have a sum of zero, i.e., $P + Q = 0$.

2. Understand the implications:

- If $P + Q = 0$, then $Q = -P$.
- The lines are perpendicular, meaning the product of their slopes is -1 .

3. Equation of the lines:

- Assume the lines have equations of the form $y = m_1x + P$ and $y = m_2x + Q$.
- Since $P = -Q$, the equations become $y = m_1x + P$ and $y = m_2x - P$.

4. Use the intersection point:

- Both lines pass through $(6,8)$, so substitute $x = 6$ and $y = 8$ into the line equations:

$$\begin{aligned} 8 &= m_1(6) + P \quad \text{and} \quad 8 = m_2(6) - P \end{aligned}$$

5. Solve for P and Q :

- From $8 = 6m_1 + P$, we get $P = 8 - 6m_1$.
- From $8 = 6m_2 - P$, we get $P = 8 - 6m_2$.
- Since $P = -Q$ and $P = -Q$ implies $P = 0$ and $Q = 0$, we find:

$$\begin{aligned} 0 &= 8 - 6m_1 \implies 6m_1 = 8 \implies m_1 = \frac{4}{3} \\ 0 &= 8 - 6m_2 \implies 6m_2 = 8 \implies m_2 = \frac{4}{3} \end{aligned}$$

(However, we need to reconsider the perpendicular condition).

6. Correct understanding of perpendicularity:

- The correct perpendicular lines are $y = \frac{4}{3}x - 8$ and $y = -\frac{4}{3}x + 8$.
- The equations are adjusted for the correct slopes.

7. Vertices of $\triangle APQ$:

- $A(6,8)$
- $P(0,-2)$
- $Q(0,2)$ (since $P = -Q$ and $P = 8$ and $Q = -8$)

8. Calculate the area of $\triangle APQ$:

- Use the formula for the area of a triangle with vertices at (x_1, y_1) , (x_2, y_2) , (x_3, y_3) :

$$\begin{aligned} \text{Area} &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ \text{Area} &= \frac{1}{2} |6(2 - 8) + 0(8 - 8) + 0(8 - 2)| \\ &= \frac{1}{2} |6(-6)| = \frac{1}{2} | -36 | = 18 \end{aligned}$$